

DATABASE DOCUMENTATION

Project Performance Improvement Toolkit

File-based inventory and logical relationships

Technical source: oli_v6_deploy.py

Updated: May 11, 2026

1. Data Model

The application does not use a relational database. The data model is file-based: Excel, pipe-delimited CSV, PyTorch tensors, and local pickle caches.

Relationships are logical and performed in memory with pandas. There are no SQL migrations, constraints, or transaction controls.

2. Data File Inventory

File	Loaded By	Use
df_complete_all_full.xlsx	load_data	Main base. Creates index_df from ID_Recomendacion, normalizes column names and dimension/subdim.
df_split_actions.xlsx	load_data	Raw base with Recommendation date; contributes year and index_df to fill missing values.
analyzed_recommendations_plans_v5.csv	load_extended_data	CSV with sep=' '. Adds additional analysis, tags, scores, and visualization metrics.
emb_Recomm_rec_cl_4.pt	load_embeddings	Recommendation embedding matrix; active startup dependency because load_embeddings runs even though the visible search is commented out.
Recommendation_RAG_Metadata.pt	load_embeddings	Metadata aligned to emb_Recomm_rec_cl_4.pt; active startup dependency.
emb_LL_ll_cl_4.pt	load_lessons_embeddings	Embeddings for learned lessons; helper not connected to an active tab.
lessons_metadata.pt	load_lessons_embeddings	Expected metadata for learned lessons; helper not connected to an active tab.
Appraisal Checklist_2025es-419.xlsx	load_appraisal_questions	rubric sheet; requires Pregunta_Realizada and uses Tema to renumber questions.
TRANSVERSAL_MATTERS	Tab 1	In-code dictionary, not an external file. Recognizes gender, non-discrimination, disability, social dialogue and tripartism,

		and environmental sustainability for the A/B rule.
Rubricas_6ago2025.xlsx	Tab 2	The active UI evaluates rubric_parteval, rubric_gender_, and rubric_TJ_TJ. Other sheets exist or are loaded in legacy code, but are not active options.
Evaluacion de sostenibilidad del proyecto_rubric_9feb26.xlsx	Tab 3	rubric sheet; active source of sustainability criteria.
Recommendations_World.xlsx	Tabs 5 and 6	World dataset for classification; can be replaced by uploaded XLSX with Recommendation description.
Frame_Recommendations_English.xlsx	Tabs 5 and 6	Reference frame; must contain texto_merged, dimension, and subdim.
embeddings_cache.pkl	EmbeddingsCache	Persistent local cache of classification embeddings for Tabs 5 and 6; not encrypted or integrity-checked.
analysis_cache.pkl	AnalysisCache	Persistent local cache of deep recommendation analyses; not encrypted or integrity-checked.

3. Logical Relationships

- df_complete_all_full.xlsx and df_split_actions.xlsx align by index_df.
- analyzed_recommendations_plans_v5.csv is merged into df by index_df when new columns exist.
- emb_Recomm_rec_cl_4.pt must stay in the same order as Recommendation_RAG_Metadata.pt for the in-memory recommendation index.
- Recommendations_World.xlsx is classified against each row of Frame_Recommendations_English.xlsx using texto_merged as the semantic definition; this flow uses numpy cosine similarity, not FAISS.
- Tabs 5 and 6 deduplicate embeddings by recommendation text, but results return to row level to preserve repeated attributes.
- Recommendation datasets can contain duplicated rows for multiple attributes; Tabs 5 and 6 show both record counts and unique recommendation counts.

4. Main Transformations

Transformation	Detail
Column normalization	Spaces and periods are replaced by underscores in load_data.
Years	Recommendation_date and Recommendation date are parsed with pandas; years before 2018 in analyzed_df are set to 2018.
Categories	prepare_additional_data standardizes categorical fields, lists/tags, and classifications when present.
FAISS	A recommendation index is built at startup. Tab 4 rebuilds an IndexFlatIP for each large-document

	RAG question, using chunks already embedded and stored in session.
Transversal matters	In Tab 1, questions detected by TRANSVERSAL_MATTERS bypass A-E and are graded with A=objective/output/activity and B=budget; Yes requires A+B, Partial requires A or B, No requires both absent.
Classification	Each unique recommendation is embedded once, normalized, and compared against the frame with numpy cosine similarity. The base assignment uses top 3; optional LLM verification reranks top 10; secondary dimensions/subdimensions are retained when similarity ≥ 0.60 .
Export	Results are written to XLSX with ordered columns and strings_to_urls disabled.

5. Minimum Fields by Flow

Flow	Minimum Required Fields
Preliminary appraisal	Appraisal Checklist: Pregunta_Realizada; Tema is recommended. Transversal questions are recognized by in-code aliases.
Specific attributes	Rubricas_6ago2025: active sheets rubric_parteval, rubric_gender_, and rubric_TJ_TJ, with Indicador, Dimension, and performance levels.
Sustainability	Active rubric: dimension, criterion, indicator, and level columns according to the rubric sheet.
Classification	Recommendations_World: Recommendation description. Frame: texto_merged, dimension, subdim.
Deep analysis	Recommendation description and Action plan; Comments improves analysis when present.
Executive summary	Recommendation description; uses Management response, Comments, and Action plan when present.

6. Backup and Data Updates

- Before replacing an Excel, CSV, or .pt tensor, keep a dated copy outside the runtime directory.
- After replacing a data file, restart the app or clear Streamlit cache.
- If embeddings are regenerated, regenerate the aligned metadata file as well; do not mix tensors and metadata from different runs.
- If the schema of Recommendations_World.xlsx or Frame_Recommendations_English.xlsx changes, validate Tabs 5 and 6 with a small sample before full processing.
- Do not manually edit embeddings_cache.pkl or analysis_cache.pkl; if corruption is suspected, delete the file with the app stopped so it can be regenerated.
- Do not assume emb_LL_ll_cl_4.pt or lessons_metadata.pt are in operational use unless the lessons function is reconnected to an active tab.